ASSIGNMENT-19.2

Task 1 – Python to Java Conversion

Task:

Convert a Python program that performs file handling operations into an equivalent Java program using AI assistance.

Instructions:

- **Prompt AI to translate a Python script that reads from and writes to a file into Java.**
- **Ensure proper handling of file streams, exceptions, and input/output formatting.**
- **Verify that the Java program produces the same file content and console output as the Python version.**
- **Test both implementations with sample input files.**

Expected Output:

A correctly translated and executable Java program that performs the same file handling operations as the original Python script.

FileHandler.java > FileHandler > main(String[])

```java
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;

public class FileHandler {
    Run | Debug
    public static void main(String[] args) {
        String data = "";

        // Read from input.txt
        try (BufferedReader reader = new BufferedReader(new FileReader(fileName: "input.txt"))
            StringBuilder content = new StringBuilder();
            String line;

            while ((line = reader.readLine()) != null) {
                content.append(line).append(str: "\n");
            }

            // Remove the last newline if present
            data = content.toString().trim();

        } catch (IOException e) {
            System.err.println("Error reading file: " + e.getMessage());
            return;
```

PYTHON TO JAVA FILE HANDLING CON...

Convert the following Python file handling program into Java.

Requirements:

- Use appropriate Java file handling classes (FileReader, BufferedReader, FileWriter, etc.)
- Handle exceptions using try-catch
- Maintain the same functionality and output formatting

Also provide:

1. Explanation of the Java code
2. Differences between Python and Java file handling
3. Sample input and output for verification

Python Code:

OUTPUT:

```
HELLO WORLD
JAVA FILE HANDLING
PROGRAMMING CONCEPTS
PROCESSING TEXT FILES
```

Task 2 – C++ to Python Function Conversion

Task:

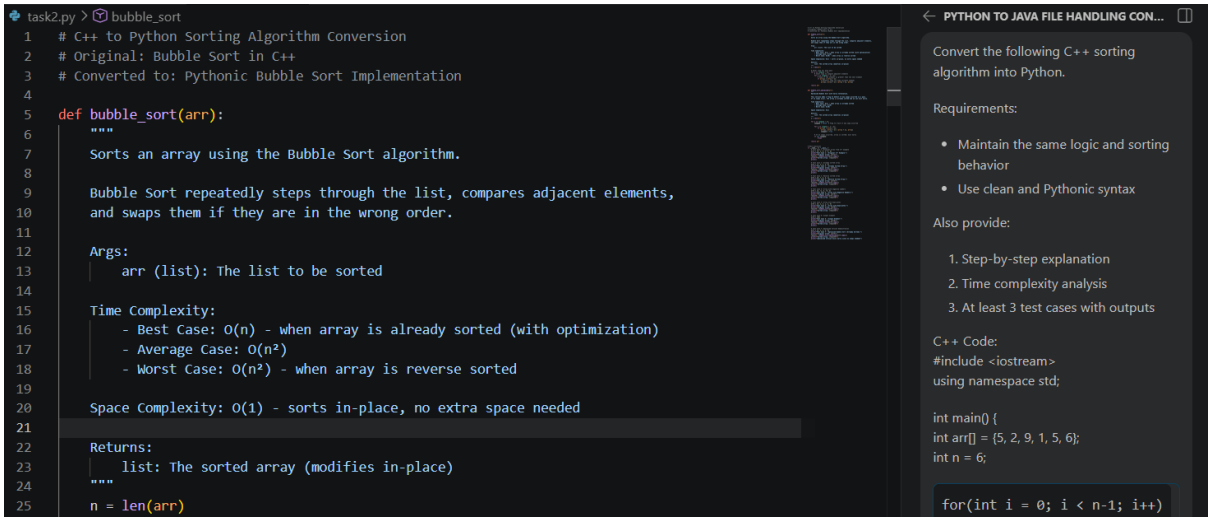
Translate a C++ program that implements a sorting algorithm into Python using AI assistance.

Instructions:

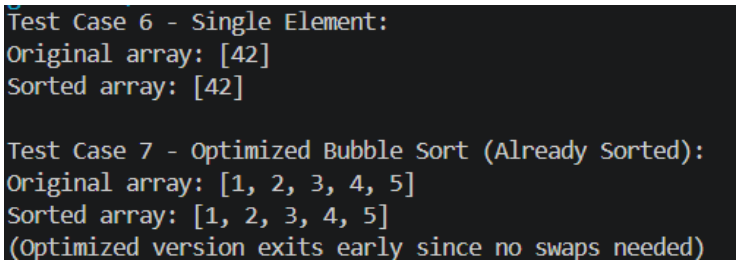
- **Provide AI with a C++ implementation of a sorting technique (e.g., Bubble Sort or Selection Sort).**
- **Ask AI to generate an equivalent Python implementation.**
- **Ensure logical correctness and validate results using multiple test cases.**
- **Confirm that the Python version maintains the same sorting behavior.**

Expected Output:

A correct and executable Python program equivalent to the given C++ sorting implementation.



OUTPUT:



Task 3 – SQL to Pandas Query

Task:

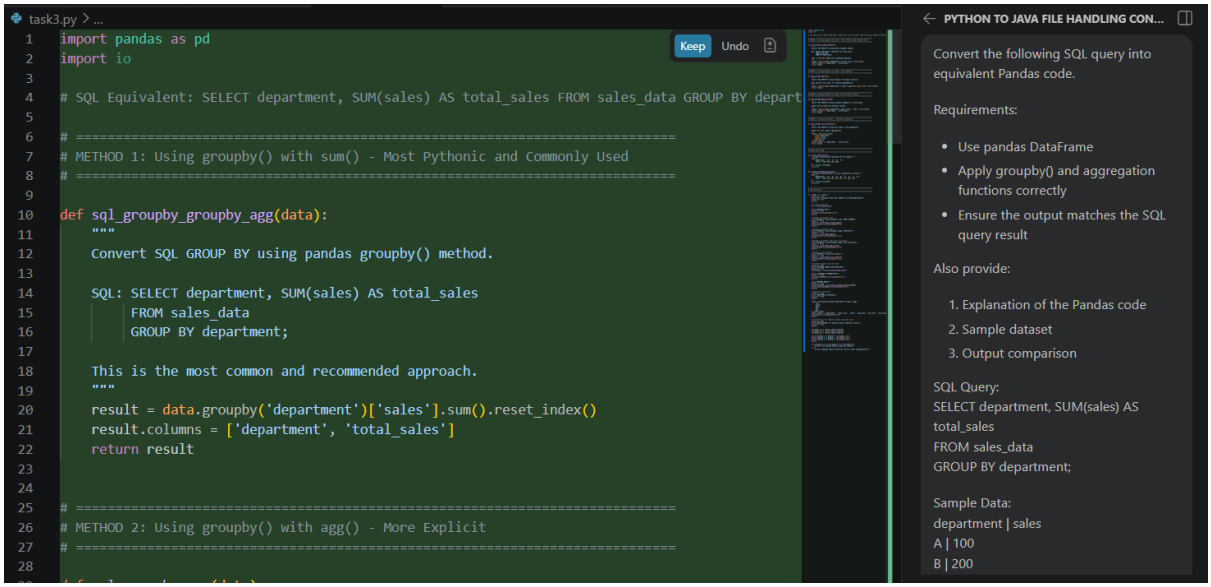
Convert a SQL query involving GROUP BY and aggregate functions into an equivalent Pandas Data Frame operation.

Instructions:

- **Provide AI with a SQL query that calculates aggregate values (e.g., total sales per department).**
- **Prompt AI to generate equivalent Pandas code using groupby() and aggregation functions.**
- **Test the generated code using a sample dataset.**
- **Verify that both SQL and Pandas outputs match.**

Expected Output:

An accurate Pandas Data Frame operation that produces results equivalent to the given SQL query with aggregation.



OUTPUT:

```
Method 1 == Method 2: True
Method 1 == Method 3: True
Method 1 == Method 4: True

✓ All methods produce identical output!
```

Task 4 – Real-Time Application: Multi-Language Snippet Converter

Scenario:

Students will select a practical real-world application requiring the same logic in two different programming languages.

Example:

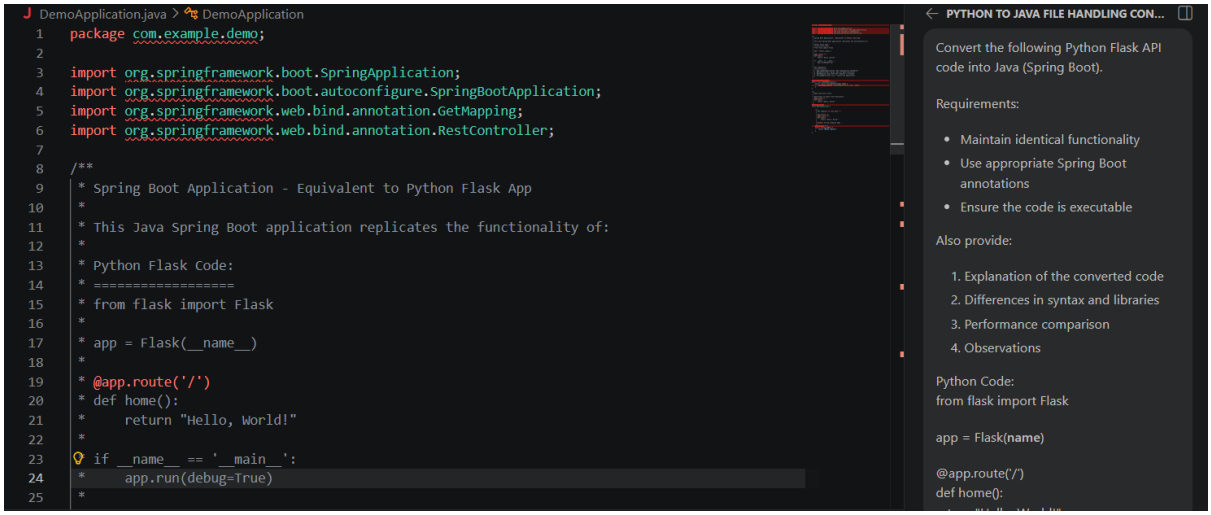
- **Converting a REST API implementation from Python (Flask) to Java (Spring Boot).**
- **Translating a data processing script from Python to C++ for performance optimization.**

Instructions:

- **Provide AI with source code in one programming language.**
- **Prompt AI to convert the code into another chosen language.**
- **Execute and test both versions to ensure identical functionality.**
- **Document differences observed in syntax, libraries, and execution behavior.**

Expected Output:

Functionally equivalent and executable code snippets implemented in two different programming languages, along with documented observations.



OUTPUT:

```
Hello, World!
```

Task -5: (Basic Function Translation (Python to C++)

Translate a simple Python function that checks whether a number is even or odd into C++ using AI assistance.

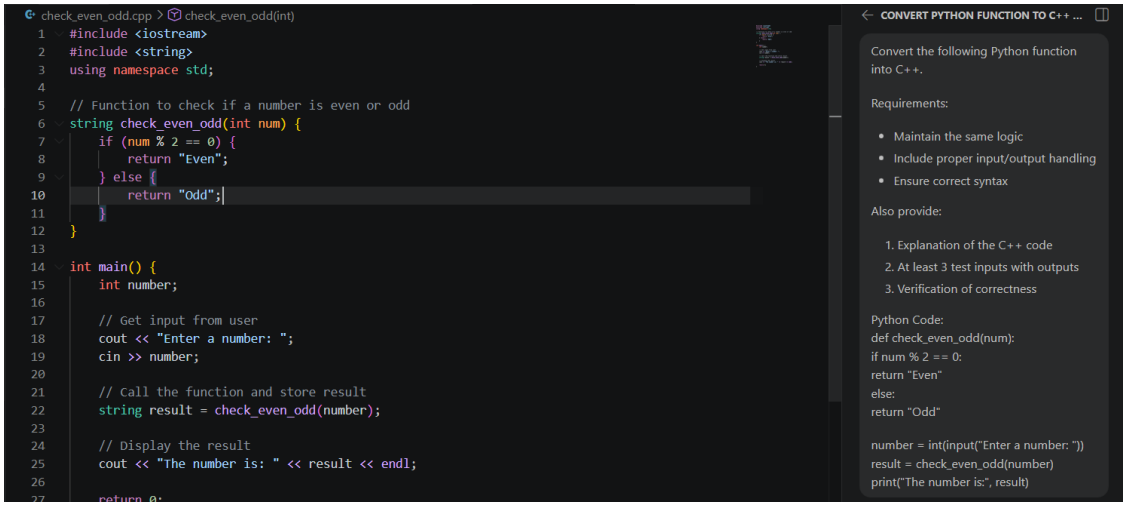
Instructions:

- **Provide AI with a basic Python function that determines if a number is even or odd.**
- **Prompt AI to generate equivalent C++ code.**
- **Ensure correct syntax, input handling, and output formatting.**
- **Compile and run the C++ program to verify it produces the same result as the Python version.**
- **Test with at least three different input values.**

Expected Output:

A simple and executable C++ program equivalent to the given Python function, correctly identifying whether a number is even or odd for all test cases.

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.



OUTPUT:

```
Enter a number: 4
The number is: Even
```